INCH-POUND
MIL-PRF-3098/104G
1 November 2017
SUPERSEDING
MIL-PRF-3098/104F
6 March 2013

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR102/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 50 MHz to 125 MHz; fifth overtone; noncontrolled; series resonance.

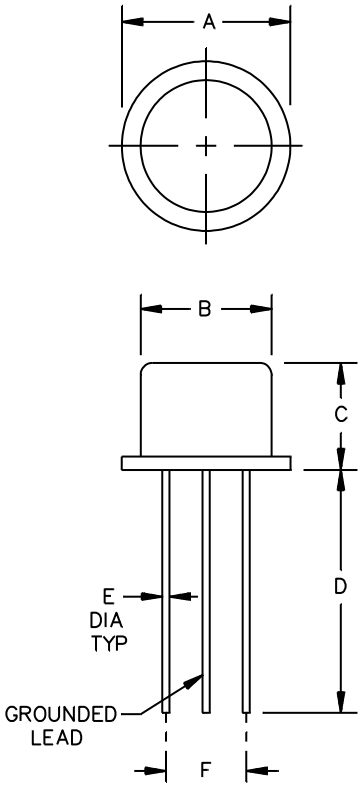


FIGURE 1. Crystal unit - CR102/U.

AMSC N/A
FSC 5955

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.420	---	10.67
B	---	.326	---	8.28
C	---	.265	---	6.73
D	.600	---	15.24	---
E	.016	.020	.406	.508
F	.192	.208	4.88	5.28

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR102/U, crystal unit - Continued.

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REQUIREMENT:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50 MHz to 125 MHz, inclusive.

Frequency tolerance: ±25 ppm.

Equivalent resistance: 50 MHz to 90 MHz: 50 ohms, maximum.
90 MHz to 125 MHz: 60 ohms, maximum.

Mode of oscillation: Fifth overtone.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 7 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Vibration: MIL-STD-202-204, test condition B.

Frequency change permitted: ±5 ppm.

Equivalent-resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204

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Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 84, 99

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2017-012)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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